AI-Assisted Screening of Candidate Pond-Based Drafting Locations in a Rural Wildland-Urban Interface Fire Response Area

Abstract

Rural wildland-urban interface fire departments often operate without hydrant networks and with incomplete inventories of static water sources. This study evaluated a multimodal artificial-intelligence workflow for screening high-resolution National Agriculture Imagery Program imagery for ponds with nearby apparent vehicle access in the approximately 25,400-ha Rist Canyon Volunteer Fire Department response area in Colorado. A baseline 500 m grid generated 1,001 RGB image chips interpreted with GPT-5.1, and two human reviewers separately labeled the same scenes. A coarser 1,000 m grid-and-chip design and Claude Sonnet 4.6 were used for sensitivity analyses. GPT-5.1 labeled 72 scenes as pond-positive, including 49 likely-draftable and 23 draftability-uncertain scenes, and produced 96 raw candidate centers that formed 88 clusters at a 100 m threshold. Pairwise pond-presence F1 scores were 0.889 and 0.820 relative to Reviewers 1 and 2, respectively and corresponding binary draftability F1 scores were 0.860 and 0.714. The 1,000 m design recovered 37 of 88 baseline clusters (42.0%), whereas Claude recovered 61 of 88 (69.3%). Because positive scenes were uncommon, high overall agreement was partly driven by the large number of negative scenes, and disagreements among positive cases remained operationally important. The workflow is best interpreted as first-pass prioritization for local review, not as a confirmed drafting-site inventory.

Keywords: wildland-urban interface; static water supply; multimodal AI; NAIP; pre-incident planning; vision-language model

1. Introduction

Wildland fire response in the wildland–urban interface (WUI) increasingly depends on spatial information that is operationally meaningful at the scale of roads, water sources, and apparatus movement. Across the United States, residential development has expanded substantially into fire-prone landscapes, increasing the number of homes and communities exposed to wildfire (Radeloff et al., 2005, 2018). This expanding exposure creates additional challenges for wildfire risk management, community preparedness, and emergency response in the WUI (Calkin et al., 2014). In these settings, wildfire risk is not defined only by fuels, weather, and topography; it is also shaped by whether responders can safely reach threatened areas, locate usable water, stage apparatus, and sustain suppression actions under changing conditions. For rural and volunteer fire departments, these operational constraints can be especially important given that formal hydrant infrastructure is often absent, access routes may be privately maintained, and local knowledge may not be represented in public geospatial datasets.

Water supply is an important constraint in wildland fire response. Fire-service standards recognize the importance of alternative water supplies in suburban and rural firefighting, including static sources such as ponds, lakes, and reservoirs (National Fire Protection Association, 2017). In wildland fire operations, drafting refers to drawing water from a static source into a pump, and water sources can be used to refill mobile water-supply apparatus such as engines and water tenders (National Wildfire Coordinating Group, n.d.). In this study, draftability refers to the apparent potential for a static water source to be accessed and used by fire-service apparatus for drafting or other water-supply operations. The presence of water alone does not establish draftability. A pond may be visible in aerial imagery but located too far from a usable road for safe and practical apparatus positioning. Conversely, a small pond located near a private driveway, ranch road, two-track, or cleared pullout may be valuable for pre-incident situational awareness even if it is absent from formal water-source inventories or public road-network datasets. This distinction between “water body” and “operationally relevant candidate water source” is central to this study.

The need for operational water-source awareness is particularly clear in the Rist Canyon Volunteer Fire Department (RCVFD) response area in Larimer County, Colorado. The RCVFD Community Wildfire Protection Plan characterizes the response area as a classic wildland–urban interface landscape with steep and rugged terrain, poor access roads, and no established water system (Rist Canyon Volunteer Fire Department, 2010). The area has also experienced two major wildfires that underscore the operational challenges associated with this landscape. The 2012

High Park Fire burned approximately 87,284 acres while the 2020 Cameron Peak Fire burned 208,913 acres and became the largest wildfire in Colorado's recorded history. The RCVFD plan notes that numerous ponds and streams provide water through drafting (Rist Canyon Volunteer Fire Department, 2010). These local conditions and the area's wildfire history illustrate a broader challenge for rural WUI response, potentially important water sources may exist across the landscape, but their operational value depends on the spatial relationship between water and apparatus access. A geospatial workflow that maps ponds alone is therefore insufficient for fire-department preplanning. What is needed is a method for identifying candidate drafting locations where visible water and likely vehicle access occur together.

Geographic information systems have long been used to support emergency response, pre-incident planning, and situational awareness by linking operationally relevant information to specific geographic features, such as structures, road networks, access points, water sources, staging areas, and incident perimeters (DeMarco, 2010). In wildfire management, spatial decision-support tools increasingly integrate landscape conditions, values at risk, potential control locations, and response-relevant information to help managers plan before incidents occur (Thompson et al., 2020). These approaches reflect a shift from static mapping toward operational geospatial intelligence. Maps are most useful when they help responders answer practical questions under time pressure. For a rural fire department, the relevant question is not simply "where are the ponds?" but rather "where are ponds that might be used as part of a water-supply strategy?" This study extends that operational logic to the problem of identifying candidate drafting locations from high-resolution aerial imagery.

Remote sensing provides a strong technical foundation for mapping surface water. Global and regional surface-water products have demonstrated the value of satellite imagery for identifying water extent, seasonality, and change over large areas (Li et al., 2022; Pekel et al., 2016). Recent studies have also used high-resolution imagery and deep learning to improve the detection of small lakes, ponds, and other surface-water features (Mullen et al., 2023). However, most surface-water mapping workflows are designed to classify water pixels or water-body objects. They generally do not evaluate whether a water feature is operationally relevant for fire apparatus access. Similarly, road-extraction research has advanced rapidly using high-resolution imagery and deep learning, but road mapping is typically treated as a separate computer-vision task rather than as part of an integrated fire-response suitability assessment (R. Liu et al., 2024). For the drafting-location problem, the relevant unit is not a water polygon or road centerline in isolation; it is a scene-level relationship among a visible water source, nearby access features, surrounding terrain and vegetation context, and the practical possibility of apparatus approach. Here, a scene-level relationship refers to how multiple features within the same image are spatially arranged and related to one another, rather than simply whether each feature is present.

High-resolution aerial imagery from the National Agriculture Imagery Program (NAIP) is well suited to this type of scene-level assessment because it provides publicly available aerial photography at fine spatial resolution across the conterminous United States (Earth Resources Observation and Science (EROS) Center, 2023). NAIP imagery commonly includes true-color red, green, and blue bands and, in newer collections, near-infrared information. While near-infrared bands are useful for spectral water detection, true-color RGB imagery approximates the visual information available to a human image analyst reviewing aerial photography. This makes RGB imagery particularly appropriate for an image-interpretation workflow where the task is not only to identify open water but also to interpret roads in a way that resembles expert visual review.

Recent advances in multimodal artificial intelligence and vision-language models create new opportunities for this type of operational remote-sensing interpretation. Remote-sensing foundation models and vision-language models have expanded the ability of AI systems to connect visual patterns with semantic descriptions, supporting tasks such as image classification, retrieval, object counting, visual question answering, and scene interpretation (F. Liu et al., 2024; Tao et al., 2025). Unlike traditional single-task classifiers, multimodal models can be prompted to evaluate compound visual criteria and return structured descriptions. This capability is especially relevant for fire-service applications where the target concept is not a standard land-cover class but an operational judgment such as whether a visible pond appears close enough to a plausible access feature to merit firefighter review as a candidate drafting location.

This study presents a reproducible workflow for identifying candidate pond-based drafting locations in the RCVFD response area using NAIP aerial imagery and multimodal AI image interpretation. The novelty of the approach is not simply automated pond detection. Instead, the workflow reframes the mapping problem as an operational screening task that combines water-source visibility with access-context interpretation. The objectives of this study were to: (1) develop a systematic, reproducible workflow for generating aerial image scenes across a rural WUI fire-response area; (2) use multimodal AI image interpretation to screen each scene for visible open water and nearby access features; (3) produce a GIS-ready point layer of candidate drafting locations for firefighter review and pre-incident planning; and (4) demonstrate how AI-assisted image interpretation can support operational situational awareness in rural fire-response areas. By focusing on candidate operational utility rather than pond presence alone, this work contributes a practical geospatial AI workflow for local wildfire preparedness and rural fire-department decision support.

2. Methods

2.1 Study area

This study was conducted within the approximately 25,400-ha RCVFD response area in the foothills of Larimer County, Colorado, USA (Figure 1). The area contains forest, rural residences, agricultural lands, scattered ponds, steep terrain, and a mixture of canyon roads, private driveways, ranch roads, and two-tracks.

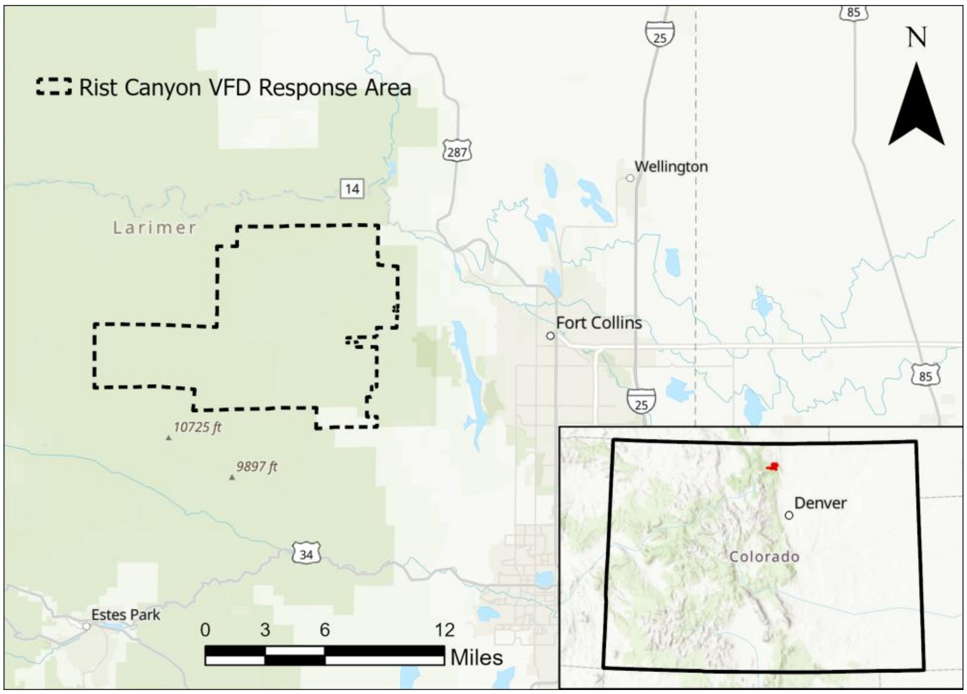


Figure 1. Location of the approximately 25,400-ha Rist Canyon Volunteer Fire Department response area in Larimer County, Colorado, USA.

2.2 Imagery source and image-chip generation

NAIP aerial imagery covering the RCVFD response area was accessed through the Microsoft Planetary Computer catalog (Earth Resources Observation and Science (EROS) Center, 2023; Microsoft Open Source et al., 2022). NAIP was selected because many ponds and access features in the study area are too small or too context dependent for interpretation in coarser regional imagery. The selected imagery was acquired on August 21, 2023, with a nominal ground-sample distance of 0.30 m.

The workflow used true-color red, green, and blue imagery. For each baseline grid point, a 500 m × 500 m chip was generated, extending 250 m in each cardinal direction from the point center (Figure 2). This extent was selected to retain local access context while keeping small ponds visually prominent. The coarser sensitivity design used both a 1,000 m grid spacing and a 1,000 m × 1,000 m chip extent; consequently, that comparison tests the combined sampling design rather than image extent alone.

Figure 2. Example 500 m × 500 m NAIP RGB image chip used for scene-level interpretation of visible water and nearby access context.

2.3 Systematic sampling grid and sensitivity design

A systematic grid with 500 m spacing was generated across the RCVFD response area and used as the baseline sampling design (Figure 3). Each grid point served as the center of one 500 m × 500 m image chip. A second grid with 1,000 m spacing was used for the sensitivity analysis, with each point serving as the center of a 1,000 m × 1,000 m chip.

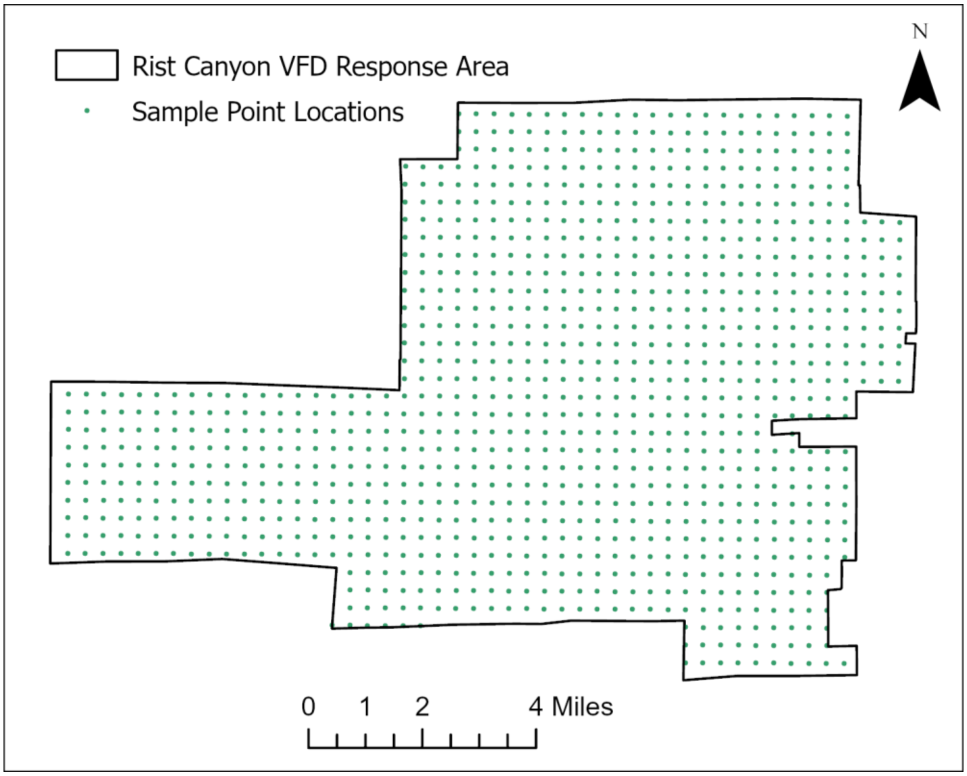


Figure 3. Systematic 500 m baseline sampling grid used to generate aerial-image chips across the RCVFD response area.

The baseline chips were not intentionally overlapped. The mapped output was therefore treated as a chip-level detection product rather than as a direct inventory of unique ponds. A pond near a chip boundary could be visible in more than one scene, while a single scene could contain multiple ponds. Each pond-positive scene was counted once in scene-level summaries, whereas each distinct pond-like feature identified within a scene could generate a separate candidate pond-center point. Spatial clustering was subsequently applied to reduce repeated detections and approximate the location of suitable ponds.

2.4 AI-based image interpretation and prompt development

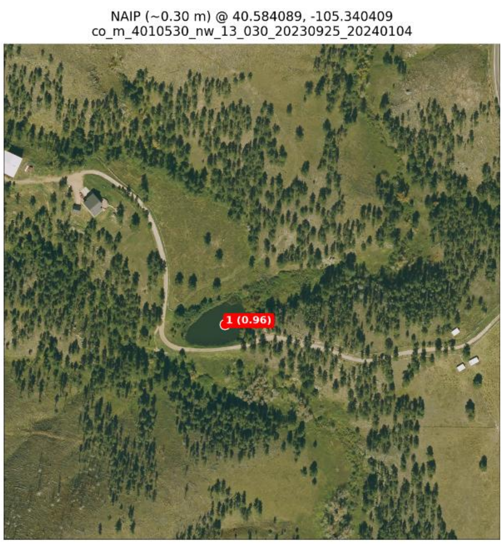
Each NAIP chip was interpreted with a multimodal AI model. OpenAI GPT-5.1 (API alias: gpt-5.1) was used as the baseline model (OpenAI, 2025). The same baseline 500 m chips and structured task were also evaluated with Claude Sonnet 4.6 (API alias: claude-sonnet-4-6) for the model-sensitivity analysis (Anthropic, 2026).

The prompt was developed through an iterative, human-guided refinement process before the full image-screening workflow was conducted. An initial version was constructed from the study objectives and the operational definitions of pond presence, apparent vehicle access, and likely draftability. The prompt was then tested on a random subset of image chips representing positive and negative conditions. Model responses were reviewed for recurring errors and inconsistencies, including confusion between water and shadows or other dark features, identification of pond-like basins without visible open water, and evaluation of pond presence without adequately considering nearby access. The instructions and decision rules were revised in response to these observed failure modes until the model responses consistently reflected the intended screening criteria and structured output format. After this development process, the prompt was finalized and applied without further modification during the full baseline analysis. The same finalized interpretation criteria and response structure were used in the model-sensitivity analysis.

The prompt instructed the model to identify visible ponds or bounded open-water bodies and then evaluate nearby roads, driveways, two-tracks, pullouts, or other apparently vehicle-accessible surfaces. It emphasized conservative interpretation and directed the model not to classify shadows, dark vegetation, roofs, tanks, wet ground, or other dark non-water features as ponds unless a bounded water feature was visible. It also directed the model not to infer likely draftability when water or access was obscured or ambiguous. The complete system instructions, image-interpretation prompt, decision rules, and structured JSON response schema are provided in Appendix B.

Each scene received standardized labels for surface-water presence, pond presence, access context, and likely draftability (Figure 4). These labels represented related but non-mutually exclusive attributes rather than a single multiclass classification. Surface-water presence indicated whether any visible open water was present, whereas pond presence indicated whether the scene contained a distinct, bounded, predominantly standing open-water body. Nearby access indicated whether vehicle-accessible surface occurred adjacent to the pond. Likely draftability represented the model's overall image-based judgment of whether the visible pond and access context warranted consideration as a potential drafting location. Surface-water presence and pond presence were recorded as yes or no, while nearby access and likely draftability were recorded as yes, no, or uncertain. Consequently, a scene could simultaneously be labeled as containing surface water, containing a pond, having nearby access, and being likely draftable.

For pond-positive scenes, the structured response could include one or more approximate pond-center coordinates, a model-reported confidence value, and a short rationale describing the visible water and access context. Responses were saved as JSON together with the source image and request metadata. Figure 4 illustrates the report structure; Appendix Figure A1 provides an additional positive example.




```
{
  "is_there_surface_water": "yes",
  "is_there_a_pond": "yes",
  "road_or_access_nearby": "yes",
  "likely_draftable": "yes",
  "pond_count": 1,
  "evidence": "An elongated dark-green feature near the image center has smooth, uniform tone and
a clear shoreline distinct from surrounding vegetation. A light-colored road directly borders its
southern edge, providing vehicle access. Size appears substantial relative to nearby house and
trees, consistent with a pond suitable for drafting.",
  "notes": "Single mid-sized pond with direct road access along its south bank; appears to be a
viable drafting candidate pending field verification.",
  "pond_points": [
    {
      "label": "pond",
      "confidence": 0.96,
      "x": 0.445,
      "y": 0.566,
      "pixel_x": 742,
      "pixel_y": 944
    }
  ]
}
```


Figure 4. Example structured AI report for a candidate pond-based drafting location. The report preserves the source chip, approximate candidate location, standardized labels, model-reported confidence, and text rationale.

2.5 Candidate definitions and screening criteria

The primary spatial output was the candidate pond-center point. A point was generated for each distinct pond-like feature returned from a pond-positive scene, whether the scene-level draftability label was likely draftable or uncertain. Each point also contained the scene-level draftability status. This distinction separates pond localization from the operational screening judgment.

A scene was labeled likely draftable when visible open water occurred near a surface that appeared capable of supporting apparatus approach. A scene was retained as draftability uncertain when a pond was visible but the access route, apparent operational scale, or other relevant context could not be judged confidently. The criteria intentionally extended beyond public road-network proximity because private and informal access routes are important in the study area and may be absent from conventional transportation datasets.

"Likely draftable" is an image-based screening label, not an operational certification. The workflow did not determine water depth, seasonal persistence, legal access, landowner permission, grade, turning radius, apparatus staging space, or firefighter safety. Final suitability requires local knowledge, landowner coordination, and field verification.

2.6 GIS processing and output generation

AI outputs were converted to spatial layers using Python 3.10.15, pandas 2.2.3, GeoPandas 1.0.1, and Shapely 2.0.6. Latitude and longitude fields were standardized and converted to point geometries in EPSG:4326. Complete interpreted-grid and candidate pond-center layers were exported for GIS review, together with tabular summaries, source chips, crop GeoTIFFs, JSON responses, and report images. After transformation to EPSG:26913, candidate points located within 100 m of one another were spatially clustered to reduce repeated detections of the same water body, and one mean-center point was retained for each cluster. Cluster centers from the baseline analysis were then matched to those from the alternative sampling-design and AI-model runs using a 150 m proximity threshold. This cross-run matching quantified how consistently candidate locations were recovered across the different workflow configurations. Additional details are provided in Section 2.8.

Mapped outputs distinguished likely-draftable from draftability-uncertain candidates and preserved links to the model response. The pond-center coordinates were approximate model-derived locations rather than manually digitized centroids. Their positional accuracy was not independently validated and should be assessed before the layer is used for site-specific operations.

2.7 Human review and agreement metrics

Two human reviewers separately classified the same 1,001 baseline 500 m image chips. The three label sets were compared pairwise by comparing Reviewer 1 versus GPT-5.1, Reviewer 2 versus GPT-5.1, and Reviewer 1 versus Reviewer 2. Neither reviewer's assessments were assumed to be completely error-free. A concordant-reviewer analysis was restricted to scenes on which both reviewers assigned the same label. Reviewers were blinded to the model outputs, which reduces the potential for model-informed assessment bias and strengthens interpretation of the agreement analysis.

Pond presence was evaluated as pond versus no pond. For the primary draftability analysis, uncertain and no labels were grouped as not likely draftable, while the original no/uncertain/yes labels were retained for the three-class summary. Pairwise comparisons used confusion matrices, overall agreement, balanced accuracy, precision, recall, F1 score, and Cohen's kappa. Simultaneous agreement among both reviewers and GPT-5.1 was summarized using complete three-way agreement and Fleiss' kappa. These statistics quantify inter-rater consistency, not field-validated operational accuracy.

2.8 Spatial clustering, sampling-design sensitivity, and model comparison

Raw candidate centers could include repeated detections of the same water body, especially near chip boundaries, and could include multiple ponds from one scene. Candidate centers were therefore clustered using a 100 m Euclidean distance threshold in EPSG:26913, with one mean-center point retained for each cluster. The 100 m threshold was a deduplication tolerance threshold.

For the sampling-design sensitivity analysis, the 500 m baseline design produced 1,001 image chips, whereas the 1,000 m design produced 252 image chips covering the same response area. The 1,000 m design used both a 1,000 m grid spacing and a 1,000 m × 1,000 m chip extent. Except for these spatial-design changes, the imagery source, GPT-5.1 model, prompt criteria, output requirements, clustering method, and spatial-matching procedures were held constant. The 500 m baseline clusters were treated as the reference candidate set. A baseline cluster was considered recovered when at least one cluster from the 1,000 m design occurred within 150 m. The reverse directional comparison quantified the proportion of 1,000 m clusters occurring within 150 m of a baseline cluster. These directional percentages do not represent one-to-one matching, and the results may vary under different clustering or matching thresholds.

For the model-sensitivity analysis, Claude Sonnet 4.6 received the identical set of 1,001 baseline 500 m image chips, prompt instructions, classification definitions, and structured output requirements used for GPT-5.1. Image preparation, response parsing, candidate extraction, 100 m clustering, and directional 150 m cross-model spatial matching were otherwise held constant. The models were compared using scene-level pond-presence classifications and binary draftability classifications. This analysis evaluated workflow sensitivity to model choice and did not establish either model as a definitive reference standard.

2.9 Workflow reproducibility

The workflow was implemented programmatically rather than through manual digitizing. Systematic grids, standardized image chips, structured prompts, machine-readable responses, archived source images, and GIS-ready exports support repeat application to other image years, model versions, chip designs, and response areas. The exact prompt, request settings, software environment, and analysis scripts are necessary for reproducibility because hosted multimodal-model behavior may change over time.

The code, prompts, and configuration files will be archived at **[INSERT PERMANENT REPOSITORY URL*!*!*!]**. Source imagery and external geospatial datasets are not redistributed and must be obtained from their original providers. Reproducing the AI interpretations also requires access to the corresponding model APIs.

3. Results

3.1 Baseline screening and candidate outputs

The baseline GPT-5.1 workflow interpreted all 1,001 image chips and identified a visible pond in 72 scenes (7.2%). It classified 49 scenes (4.9%) as likely draftable and retained 23 (2.3%) as draftability uncertain. Visible road or access context was identified in 64 scenes overall (6.4%). Within the 72 pond-positive scenes, access was labeled yes in 62 (86.1%), uncertain in six (8.3%), and no in four (5.6%). The baseline scene and point outputs are summarized in Table 1.

Table 1. Baseline GPT-5.1 scene-level classifications and candidate point outputs for the 500 m design. Candidate centers and clusters are point-based outputs and are not percentages of image chips.

Output metric	Count	Percent of 1,001 chips
Total image chips interpreted	1,001	100.0%
Pond-positive scenes	72	7.2%
Road/access visible (all scenes)	64	6.4%
Likely-draftable scenes	49	4.9%
Draftability-uncertain scenes	23	2.3%
Raw candidate pond-center points	96	Point output
Unique candidate clusters (100 m)	88	Clustered output

The 72 pond-positive scenes yielded 96 raw candidate pond-center points because 55 scenes contained one point, 10 contained two, and seven contained three. After 100 m clustering, the baseline contained 88 candidate clusters. Of the 96 raw centers, 72 were associated with likely-draftable scenes and 24 with uncertain scenes. Model-reported point confidence averaged 0.918 (median 0.930; range 0.78–0.98), with means of 0.943 for likely-draftable points and 0.842 for uncertain points. These values are uncalibrated self-reports and should not be interpreted as probabilities of correctness. The spatial distribution of the raw candidates is shown in Figure 5.

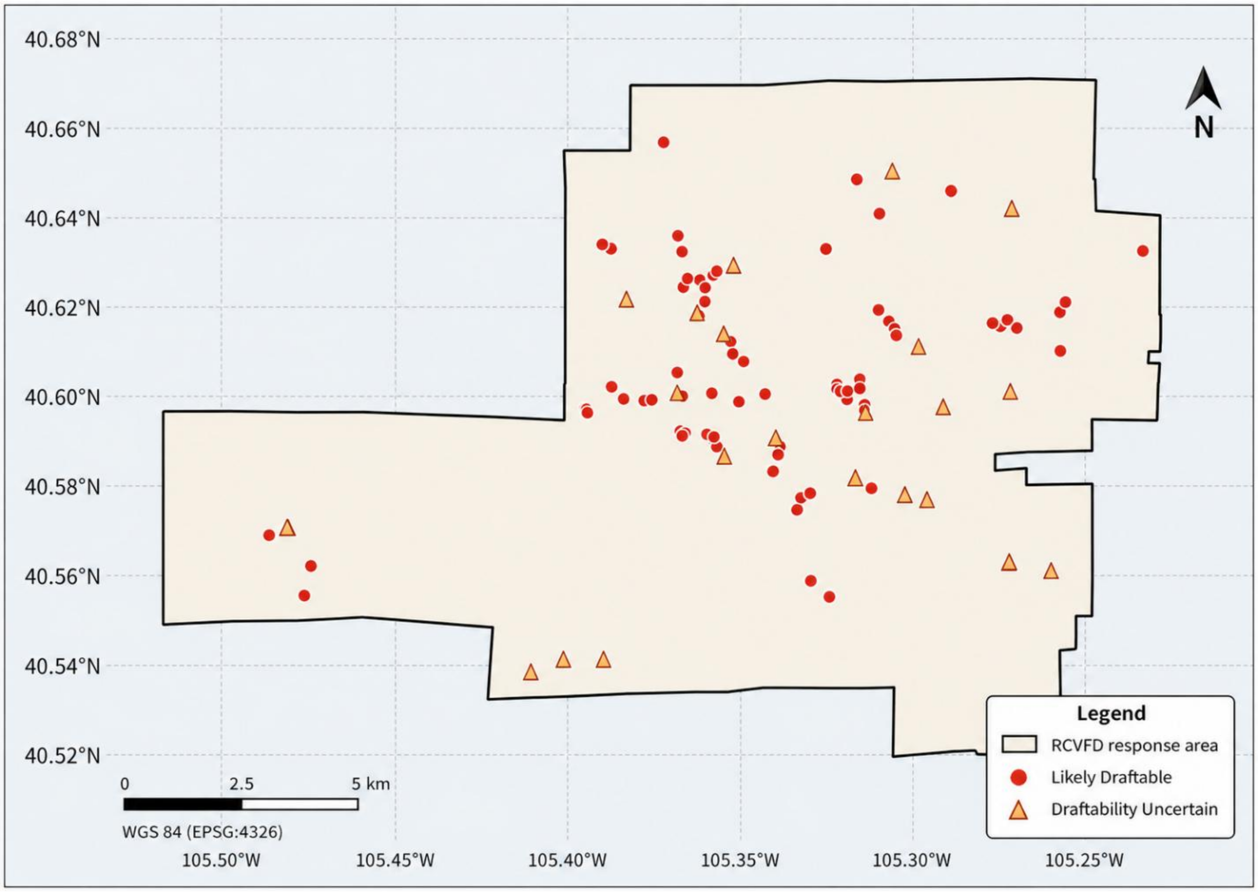


Figure 5. Spatial distribution of the 96 raw GPT-5.1 candidate pond-center points from the baseline 500 m design. Symbols distinguish likely-draftable and draftability-uncertain scenes. Points are screening outputs rather than field-verified drafting sites.

3.2 Agreement among two reviewers and GPT-5.1

Positive classifications were relatively uncommon and varied among raters. Pond-positive prevalence was 72 of 1,001 scenes (7.2%) for Reviewer 1, 89 (8.9%) for Reviewer 2, and 72 (7.2%) for GPT-5.1. For likely-draftable status, Reviewer 1 assigned 58 yes and seven uncertain labels, Reviewer 2 assigned 77 yes and nine uncertain labels, and GPT-5.1 assigned 49 yes and 23 uncertain labels. These distributions indicate that Reviewer 2 classified more scenes positively, whereas GPT-5.1 used the uncertain draftability label more frequently.

Overall agreement was high, although balanced accuracy and F1 provide more informative measures because positive scenes were uncommon. GPT-5.1 agreed more closely with Reviewer 1 than with Reviewer 2 for both pond presence and binary draftability. For pond presence, GPT-5.1 achieved balanced accuracy and F1 values of 0.940 and 0.889 relative to Reviewer 1, compared with 0.867 and 0.820 relative to Reviewer 2. For binary draftability, the corresponding values were 0.895 and 0.860 relative to Reviewer 1 and 0.790 and 0.714 relative to Reviewer 2. Complete three-way agreement was 96.6% for pond presence (Fleiss' kappa = 0.842) and 96.3% for binary draftability (Fleiss' kappa = 0.786). When no, uncertain, and yes were retained as separate draftability classes, complete agreement was 95.3% and Fleiss' kappa was 0.756.

When the analysis was restricted to scenes on which both reviewers agreed, GPT-5.1 achieved F1 scores of 0.919 for pond presence and 0.865 for binary draftability. These estimates exclude rather than adjudicate human disagreements. Selected pairwise agreement statistics are provided in Table 2, and the corresponding confusion matrices are shown in Figure 6. The confusion matrices show that disagreement was concentrated primarily in scenes classified as likely draftable by a reviewer but as not likely draftable by GPT-5.1, particularly in the comparison with Reviewer 2.

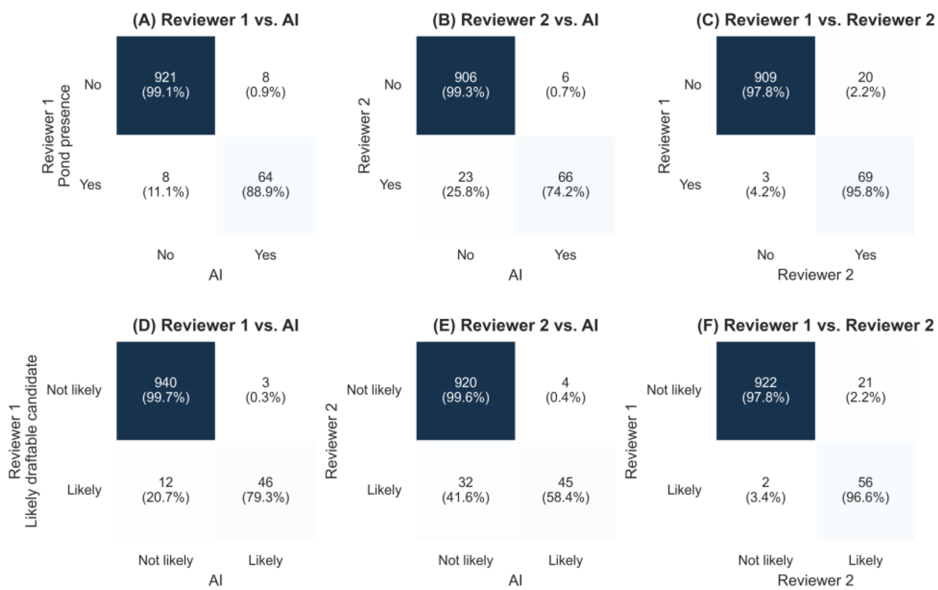


Figure 6. Pairwise confusion matrices comparing human reviewers and GPT-5.1 for pond presence and likely draftability across 1,001 baseline 500 m image chips. Panels (A–C) show binary pond-presence classifications for Reviewer 1 versus AI, Reviewer 2 versus AI, and Reviewer 1 versus Reviewer 2, respectively. Panels (D–F) show the corresponding comparisons for binary likely-draftability classifications, with "not likely" combining scenes classified as no or uncertain. Cell values report the number of image chips, with percentages calculated within each row.

Table 2. Selected pairwise agreement metrics for pond presence and binary likely-draftable status. The arrow identifies the descriptive reference and comparator used for balanced accuracy and F1; it does not imply an error-free reference. Concordant-reviewer rows include only scenes on which both reviewers agreed.

Outcome	Reference → comparator	Agreement	Balanced accuracy	F1	Cohen's kappa
Pond presence	Reviewer 1 → GPT-5.1	98.4%	0.940	0.889	0.880
Pond presence	Reviewer 2 → GPT-5.1	97.1%	0.867	0.820	0.804
Pond presence	Reviewer 1 → Reviewer 2	97.7%	0.968	0.857	0.845
Pond presence	Concordant reviewers → GPT-5.1	98.9%	0.947	0.919	0.912
Likely draftable	Reviewer 1 → GPT-5.1	98.5%	0.895	0.860	0.852
Likely draftable	Reviewer 2 → GPT-5.1	96.4%	0.790	0.714	0.696
Likely draftable	Reviewer 1 → Reviewer 2	97.7%	0.972	0.830	0.818
Likely draftable	Concordant reviewers → GPT-5.1	98.6%	0.900	0.865	0.858

3.3 Sampling-design and model sensitivity

The 1,000 m grid-and-chip design identified 30 pond-positive scenes, including 22 classified as likely draftable. The 1,000 m grid-and-chip design reduced raw candidate centers from 96 to 58 and 100 m candidate clusters from 88 to 54. Thirty-seven of 88 baseline clusters (42.0%) had a 1,000 m cluster within 150 m, whereas 36 of 54 clusters from the 1,000 m design (66.7%) occurred within 150 m of a baseline cluster. Thus, 51 baseline clusters (58.0%) were not recovered under the selected matching rule. Because grid spacing and chip extent changed together, this difference cannot be attributed solely to scene scale or object visibility.

On the same 1,001 baseline 500 m scenes, Claude Sonnet 4.6 identified 59 pond-positive and 37 likely-draftable scenes, compared with 72 and 49 for GPT-5.1. Pond-presence agreement was 97.3% (Cohen's kappa = 0.780), with 52 shared positives, 20 GPT-only positives, and seven Claude-only positives. Binary draftability agreement was 97.8% (Cohen's kappa = 0.733), with 32 shared positives, 17 GPT-only positives, and five Claude-only positives.

Claude produced 71 raw centers and 69 clusters. Sixty-one of 88 GPT-5.1 baseline clusters (69.3%) had a Claude cluster within 150 m, and 61 of 69 Claude clusters (88.4%) had a GPT-5.1 baseline cluster within 150 m. Model choice therefore changed positive-case completeness despite high overall scene agreement. Table 3 summarizes the sampling-design and model-sensitivity results.

Table 3. Sampling-design and model-sensitivity summary. Candidate centers were clustered within 100 m in EPSG:26913, and cross-run recovery used a 150 m nearest-cluster rule. Directional recovery percentages do not imply one-to-one matching.

Metric	GPT-5.1, 500 m	GPT-5.1, 1,000 m design	Claude Sonnet 4.6, 500 m
Image chips interpreted	1,001	252	1,001
Pond-positive scenes	72	30	59
Likely draftable scenes	49	22	37
Raw candidate pond-center points	96	58	71
Unique candidate clusters (100 m)	88	54	69
Baseline clusters recovered within 150 m	Reference	37 of 88 (42.0%)	61 of 88 (69.3%)
Comparison clusters within 150 m of baseline	Reference	36 of 54 (66.7%)	61 of 69 (88.4%)

4. Discussion

This study demonstrates that multimodal AI can provide a useful and repeatable screening approach for identifying candidate pond-based drafting locations from high-resolution aerial imagery. Overall, the baseline GPT-5.1 interpretation was broadly consistent with both independent human reviewers, and the Claude sensitivity run produced a substantially overlapping candidate set. Agreement was strongest for the large number of image chips that did not contain visible candidate water sources. The model was not perfect, and human reviewers still identified some missed drafting opportunities, but the AI produced results that were close enough to human interpretation to be operationally useful as a first-pass screening tool. The main advantage was not that the AI replaced human judgment, but that it rapidly and consistently reduced a large set of image chips into a smaller set of candidate locations for review. For rural fire departments with limited time, staffing, GIS capacity, and formal water-source inventories, this type of automated workflow can make pre-incident planning more efficient by directing local expertise toward the most relevant locations.

A key strength of the workflow was that the AI responses were interpretable. The model did not just return a label indicating whether a pond was present; it also provided text-based rationale that could be reviewed by a human analyst. For example, in one positive case, the model identified “a distinct dark, smooth, irregularly rounded feature” in the lower-left quadrant with “clear grassy banks and tree-lined edges” and noted that a dirt road and open vehicle area provided direct access to the pond (Appendix A1). This rationaleclosely follows the prompt design, which required both visible open water and a bounded shoreline before a feature could be classified as a pond. This also illustrates that the workflow output was not fixed or generic; instead, the structure and content of the AI response could be tailored through prompt engineering to emphasize features most relevant to the operational problem, such as open water, shoreline definition, false-positive screening, access, and apparent draftability. The model also evaluated potential false positives, noting that the feature was “distinct from tree shadows and vegetation” and “not rectangular or structural.” This is important because one of the major challenges in RGB aerial imagery is that shadows, roofs, wet ground, vegetation, and dark terrain features can resemble water. The AI’s explanation shows that it was applying the intended visual screening logic by first identifying open water, then comparing it against likely non-water explanations, and then evaluating whether visible access and apparent size supported a drafting-related interpretation.

The draftability classification also reflected the structure of the prompt rather than a simple pond-detection decision. In the same example, the AI did not classify the location as likely draftable based only on the presence of water. Instead, it connected the draftability label to visible access and apparent pond size, stating that “a dirt road and open vehicle area lead directly to the nearby building and continue adjacent to the pond” and that the pond “appears large enough and directly road-accessible for potential drafting.” (Appendix A1). This rationale is consistent with the central goal of the workflow which is to identify candidate operational water sources rather than simply mapping ponds. The mapped output should therefore be understood as a scene-level interpretation of water, access, and context. This is an important distinction from conventional water-body mapping because the relevant feature for fire preplanning is not just a pond, but a pond that appears close enough to access.

The main limitations observed in the AI results appear to be closely tied to prompt design and image-chip structure. Some candidate ponds were likely missed because each 500 m image chip was interpreted independently and conservatively. When ponds occurred near the edge of a chip, only part of the water body was visible, which could make the feature appear smaller, less clearly bounded, or more ambiguous than it would in a full-scene view. Because the prompt instructed the model to avoid uncertain detections and to return points only for clearly visible open-water ponds, partially visible edge features were more likely to be excluded rather than classified as ponds. This does not necessarily indicate that the model was incapable of identifying those ponds. Instead, it shows that chip boundaries and conservative prompt instructions can influence detection outcomes. Future implementations could reduce this issue by using overlapping image chips or by adding an explicit edge-feature rule to the prompt.

A common difference between the human review and the AI interpretation was that the AI often recognized visible ponds and access features but still assigned a conservative draftability label. The most common example of this pattern was language like “appears draftable-scale but size and depth uncertain from imagery” (Appendix A3). In these cases, the AI generally agreed that a pond was present and often noted clear access, such as a road that “directly borders” the pond. However, the model stopped short of assigning a confident likely-draftable label because the prompt emphasized that operational suitability should not be inferred unless the feature appeared large enough and usable from imagery alone. Human reviewers were often more willing to classify the same features as draftable based on visual judgment of the pond-access relationship. This makes the disagreement meaningful in that the AI was not usually failing to recognize ponds, but rather applying a stricter threshold for draftability than human reviewers.

This pattern also suggests that the "uncertain" class should not be interpreted only as model error. In an operational setting, uncertain candidate locations may represent a useful second-priority review category. A pond labeled uncertain may still be valuable if it is located in a strategically important area, near homes, near a long access route, or in a portion of the response area with limited known water supply. The uncertain label therefore provides a way to preserve potentially useful locations without overstating confidence. For fire-department preplanning, this may be preferable to a binary classification that forces every site into either likely draftable or not draftable.

The sensitivity test showed that image-chip scale strongly affected candidate recovery. The 1000 m image chips reduced the number of model calls and processing requirements, but they recovered fewer candidate pond locations than the 500 m workflow. This likely occurred because small ponds occupied a smaller proportion of the image scene in the larger chips and were therefore less visually obvious to the model. Even when the underlying NAIP resolution remained the same, the visual interpretation task changed because the model had to evaluate more landscape context at once. For this application, the 500 m chip size provided a better balance between context and object visibility. This finding is important because it shows that multimodal AI performance depends not only on image resolution, but also on scene framing, object size, and the scale at which the prompt asks the model to provide rationale.

The model-sensitivity analysis similarly showed that the workflow's conclusions were not tied to one model, but that the exact candidate inventory was model dependent. GPT-5.1 and Claude Sonnet 4.6 agreed on more than 97% of scene-level pond and binary draftability classifications, yet the agreement and cluster-recovery results revealed meaningful differences among the rarer positive cases. Claude produced fewer pond-positive and likely-draftable scenes, and it recovered about 69% of the GPT-5.1 baseline clusters. Thus, changing the multimodal model had a smaller effect than doubling the image-chip extent, but it still altered which locations would advance to firefighter review. Operational deployments should therefore retain model identifiers and prompts, evaluate model changes before replacing a workflow, and preserve human review for the positive and uncertain candidates.

The results also show why raw AI point outputs should be interpreted as image-chip-level candidate detections rather than a finalized pond inventory. Some image chips contained more than one pond, and ponds near chip boundaries could potentially be detected from more than one neighboring scene. As a result, mapped pond-center points may exceed the number of unique water bodies on the landscape. For operational use, the raw AI detections should be followed by spatial clustering, duplicate removal, and local review before being treated as a unique water-source layer. This distinction is important because the workflow was designed as a screening and prioritization tool, not as a final authoritative inventory of all ponds or confirmed drafting sites.

Several improvements could strengthen future versions of the workflow. First, the prompt could be revised to handle partially visible edge ponds more explicitly, including instructions for how to classify features that are cut off by the chip boundary. Second, overlapping image chips could reduce missed detections caused by edge effects. Third, a multi-scale approach could combine smaller chips for pond detection with larger chips for broader access and terrain context. Fourth, RGB image interpretation could be supplemented with additional geospatial layers such as near-infrared imagery, water indices, slope, or road networks.

Overall, the AI workflow performed best as a transparent and conservative system. It was able to apply structured visual rationale across a large number of image chips, identify visible ponds, assess nearby access context, and produce GIS-ready candidate points for review. The main disagreements with human interpretation were not random and were concentrated around understandable sources of uncertainty. This suggests that, in this example, multimodal AI can be valuable for rural WUI fire preplanning when it is used in the correct role which is not as a replacement for firefighter judgment, but as a scalable screening tool that helps local experts find and prioritize candidate water sources more efficiently.

Data and code availability

The analysis code, prompts, and configuration files will be archived at **[INSERT PERMANENT REPOSITORY URL*!*!*!]**. NAIP imagery and other external data must be obtained from their original providers. Raw API responses and model/request metadata should be retained with the archived workflow because hosted model aliases may change.

References

Anthropic. (2026, February). *Claude Sonnet 4.6*. https://www.anthropic.com/claude/sonnet

Calkin, D. E., Cohen, J. D., Finney, M. A., & Thompson, M. P. (2014). How risk management can prevent future wildfire disasters in the wildland-urban interface. *Proceedings of the National Academy of Sciences, 111*(2), 746–751.

DeMarco, D. L. (2010). *Using Geographic Information Systems for Pre-Incident Planning* [Applied Research Project, Executive Fire Officer Program]. National Fire Academy. https://apps.usfa.fema.gov/pdf/efop/efo45087.pdf

Earth Resources Observation and Science (EROS) Center. (2023). *National Agriculture Imagery Program: 2003–present*. U.S. Geological Survey. https://doi.org/10.5066/F7QN651G

Li, J., Ma, R., Cao, Z., Xue, K., Xiong, J., Hu, M., & Feng, X. (2022). Satellite detection of surface water extent: A review of methodology. *Water*, *14*(7), 1148.

Liu, F., Chen, D., Guan, Z., Zhou, X., Zhu, J., Ye, Q., Fu, L., & Zhou, J. (2024). Remoteclip: A vision language foundation model for remote sensing. *IEEE Transactions on Geoscience and Remote Sensing*, *62*, 1–16.

Liu, R., Wu, J., Lu, W., Miao, Q., Zhang, H., Liu, X., Lu, Z., & Li, L. (2024). A review of deep learning-based methods for road extraction from high-resolution remote sensing images. *Remote Sensing*, *16*(12), 2056.

Microsoft Open Source, McFarland, M., Emanuele, R., Morris, D., & Augspurger, T. (2022). *microsoft/PlanetaryComputer: October 2022* (Version 2022.10.28) [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.7261897

Mullen, A. L., Watts, J. D., Rogers, B. M., Carroll, M. L., Elder, C. D., Noomah, J., Williams, Z., Caraballo-Vega, J. A., Bredder, A., Rickenbaugh, E., & others. (2023). Using high-resolution satellite imagery and deep learning to track dynamic seasonality in small water bodies. *Geophysical Research Letters*, *50*(7), e2022GL102327.

National Fire Protection Association. (2017). *NFPA 1142: Standard on Water Supplies for Suburban and Rural Fire Fighting*. National Fire Protection Association.

National Wildfire Coordinating Group. (n.d.). *Draft: NWCG Glossary of Wildland Fire, PMS 205*. Retrieved July 15, 2026, from https://www.nwcg.gov/node/5056100

OpenAI. (2025). *GPT-5.1 Model*. https://developers.openai.com/api/docs/models/gpt-5.1

Pekel, J.-F., Cottam, A., Gorelick, N., & Belward, A. S. (2016). High-resolution mapping of global surface water and its long-term changes. *Nature*, *540*(7633), 418–422.

Radeloff, V. C., Hammer, R. B., Stewart, S. I., Fried, J. S., Holcomb, S. S., & McKeefry, J. F. (2005). The wildland–urban interface in the United States. *Ecological Applications*, *15*(3), 799–805.

Radeloff, V. C., Helmers, D. P., Kramer, H. A., Mockrin, M. H., Alexandre, P. M., Bar-Massada, A., Butsic, V., Hawbaker, T. J., Martinuzzi, S., Syphard, A. D., & others. (2018). Rapid growth of the US wildland-urban interface raises wildfire risk. *Proceedings of the National Academy of Sciences*, *115*(13), 3314–3319.

Rist Canyon Volunteer Fire Department. (2010). *Community Wildfire Protection Plan*. Rist Canyon Volunteer Fire Department. https://static.colostate.edu/client-files/csfs/documents/RistCanyon_CWPP.pdf

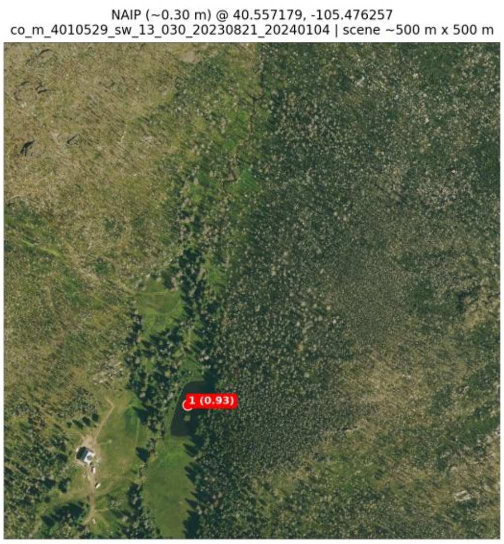
Tao, L., Zhang, H., Jing, H., Liu, Y., Yan, D., Wei, G., & Xue, X. (2025). Advancements in vision–language models for remote sensing: Datasets, capabilities, and enhancement techniques. *Remote Sensing*, *17*(1), 162.

Thompson, M. P., Gannon, B. M., Caggiano, M. D., O’Connor, C. D., Brough, A., Gilbertson-Day, J. W., & Scott, J. H. (2020). Prototyping a geospatial atlas for wildfire planning and management. *Forests*, *11*(9), 909.

Appendix A. Example AI interpretations

The appendix provides examples of the image evidence and structured model responses used in the workflow. The examples illustrate a likely-draftable candidate, a conservative no-pond classification, and a draftability-uncertain candidate. Model-generated text is included as an auditable output, not as evidence of transparent internal rationale.

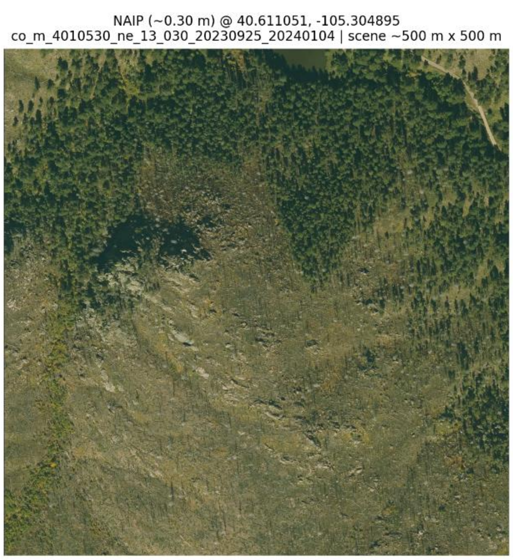
Appendix A1. Likely-draftable pond candidate




```
{
  "is_there_surface_water": "yes",
  "is_there_a_pond": "yes",
  "road_or_access_nearby": "yes",
  "likely_draftable": "yes",
  "pond_count": 1,
  "evidence": "Near the lower-left quadrant is a distinct dark, smooth, irregularly rounded
feature with clear grassy banks and tree-lined edges, unlike surrounding shadows or vegetation. A
dirt road and open vehicle area lead directly to the nearby building and continue adjacent to the
pond, indicating clear access and a non-tiny water body.",
  "notes": "Single accessible pond near structures in southwest quadrant; appears large enough and
directly road-accessible for potential drafting.",
  "pond_points": [
    {
      "label": "pond",
      "confidence": 0.93,
      "x": 0.37,
      "y": 0.73,
      "pixel_x": 617,
      "pixel_y": 1217
    }
  ]
}
```


Appendix Figure A1. Positive candidate in which the model identified visible open water, a bounded shoreline, and nearby apparent access.

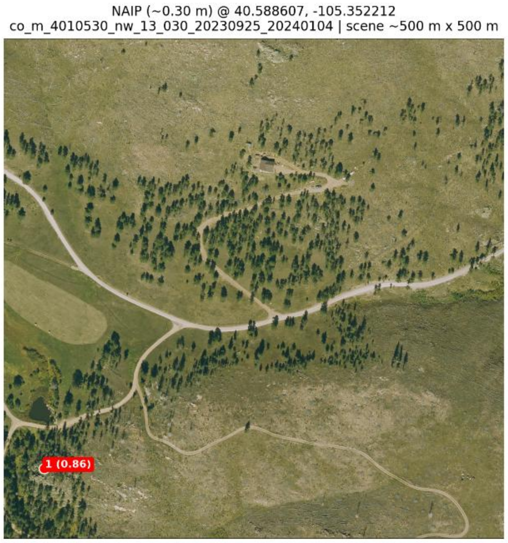
Appendix A2. Conservative no-pond classification




```
{
  "is_there_surface_water": "no",
  "is_there_a_pond": "no",
  "road_or_access_nearby": "no",
  "likely_draftable": "no",
  "pond_count": 0,
  "evidence": "Scene shows rocky, sparsely forested terrain with tree shadows and textured
ground. Dark areas within the forest and rocks match surrounding shadow patterns and lack
smooth, uniform interiors or clear shorelines. No distinct, bounded open-water surface is
visible anywhere in the crop.",
  "notes": "No clearly identifiable ponds or surface water sources for drafting are visible in
this image.",
  "pond_points": []
}
```


Appendix Figure A2. Negative scene in which the model did not identify a clearly bounded pond or open-water source and therefore returned no candidate center.

Appendix A3. Draftability-uncertain candidate




```
{
  "is_there_surface_water": "yes",
  "is_there_a_pond": "yes",
  "road_or_access_nearby": "yes",
  "likely_draftable": "uncertain",
  "pond_count": 1,
  "evidence": "Single dark, smooth, oval feature at lower-left with clear shoreline distinct
from tree shadows and vegetation. Interior is uniform and water-like, unlike textured ground
elsewhere. A road directly borders its northern edge, providing clear vehicle access. No other
similar water-like features appear in the scene.",
  "notes": "One small pond at lower-left adjacent to road; appears draftable-scale but size and
depth uncertain from imagery.",
  "pond_points": [
    {
      "label": "pond",
      "confidence": 0.86,
      "x": 0.08,
      "y": 0.86,
      "pixel_x": 133,
      "pixel_y": 1434
    }
  ]
}
```


Appendix Figure A3. Uncertain classification in which a pond and nearby apparent access were visible, but the model did not assign a confident likely-draftable label from imagery alone.

Appendix B. Multimodal AI prompt and response format

Appendix B1. System instruction

The following system instruction was supplied to the multimodal model for each image interpretation:

You are a remote sensing analyst specializing in aerial imagery interpretation for wildfire water-source screening. Use only what is directly visible in the image. Assume NAIP is about 30 cm ground-sample distance. Be conservative and prefer false negatives over false positives. Carefully distinguish true open water from shadow, vegetation, wet ground, mud, dark soil, tanks, roofs, and structures. Evaluate road or vehicle access separately from pond presence. If a visible road, driveway, turnout, or vehicle-access area lies immediately adjacent to a clearly visible pond, mark road_or_access_nearby as yes. If a clearly visible pond has direct visible access and is not obviously tiny, likely_draftable may be yes. Only return pond-center points for clearly visible pond or open-water features. Each point must represent the approximate center of a clearly visible pond. Coordinates must be normalized from 0 to 1. If there is uncertainty, do not return a point. Return only valid JSON with the exact keys requested.

Appendix B2. Image-interpretation prompt

The following user prompt was supplied with each baseline 500 m × 500 m NAIP image chip:

Analyze this NAIP RGB aerial image at approximately 0.30 m ground-sample distance. Use only what is visible in the image. Do not assume anything not directly observable.

This task must be conservative. False positives are worse than false negatives. The primary goal is to identify visible open-water ponds that could plausibly be wildfire drafting sources.

Return only valid JSON.

Context

This image is a fixed-area crop of approximately 500 m × 500 m. The goal is to identify true visible open surface water and specifically ponds or standing water bodies that might serve as potential drafting sources.

Whole-scene context rule

- Do not judge a candidate feature in isolation.
- Compare the candidate with the rest of the image before deciding that it is water.
- Use the whole scene to determine whether a dark area is more consistent with shadow, terrain shading, tree shadow, rock shadow, or other non-water dark features that appear elsewhere in the image.
- If similar dark shapes, tones, or textures occur throughout the scene in obvious shadows or shaded terrain, prefer no for pond presence.

Definitions

Surface water: Visible open water such as a pond, lake, reservoir, stream reach with visible water, canal holding water, or wetland or open-water patch.

Water-surface appearance rule:

- Open water usually appears smoother and more internally uniform than mud, grass, sediment, or disturbed ground.
- If the interior shows mottled texture, vegetation patches, hoof-disturbed mud, exposed sediment, or land-like texture across most of the basin, do not classify it as open water.

Pond: A distinct, bounded, mostly standing open-water body with a visible open-water surface and visible edges or shoreline.

Pond basin or impoundment footprint: A depression, stock tank, basin, or pond-shaped feature that may be dry, muddy, vegetated, only faintly damp, or may contain too little visible water to matter operationally. A pond basin is not the same as visible open water.

Draftable pond: A visible open-water pond that appears, from imagery alone, to be plausibly usable as a wildfire drafting source. Positive evidence includes visible open water, a distinct shoreline, nearby road or vehicle access, and a feature that is not obviously too small. This is only an image-based screening judgment and does not confirm depth, volume, bank stability, legal access, seasonal persistence, or actual field operability.

Manmade rectangular-feature exclusion rules

- Do not consider anything perfectly square or perfectly rectangular to be a pond unless unmistakable open water is clearly visible.
- Perfect geometric shapes are strong negative evidence for ponds in this task and should usually be treated as manmade features rather than open-water ponds.
- Small dark rectangular or square features near houses, sheds, driveways, pads, or developed areas should not be classified as ponds unless clear open water is unmistakably visible.
- Roofs, sheds, covered tanks, liners, tarps, equipment pads, shadowed structures, and other manmade site features can appear dark and water-like from overhead.
- A neat rectangular feature is more likely to be a manmade object than a natural or draftable pond unless there is strong visible evidence of true open water.
- If a feature is adjacent to buildings or a maintained homesite, be especially cautious and prefer no unless water is obvious.

Shadow-comparison rule

- Before classifying a feature as a pond, compare it with other dark areas in the image.
- If the candidate has similar darkness, texture, edge quality, or orientation as nearby shadows from trees, rocks, cliffs, buildings, or terrain, do not classify it as a pond.
- If the dark feature blends gradually into surrounding shadow or appears connected to shadowed terrain, choose no for pond presence.

Scene-consistency rule

- A true pond should remain visually distinct from the broader shadow pattern of the scene.
- If the feature can be explained by the same lighting and shadow behavior seen elsewhere in the image, it is not strong evidence for water.
- Use surrounding illumination, terrain, and nearby shadows to interpret ambiguous dark features conservatively.

Contextual shadow rule for trees and rock

- In wooded, rocky, or mountainous scenes, many dark features are caused by tree shadow, terrain shadow, or rock relief.
- If a candidate dark feature resembles the shadow behavior of nearby trees, ridges, boulders, or slopes, do not classify it as a pond unless open water is unmistakable.

Important interpretation rules

1. Darkness alone is not evidence of water.
2. Do not confuse shadow, vegetation, wet ground, mud, burn scar, terrain shading, roofs, tanks, troughs, containers, disturbed pads, equipment, or structures with ponds.
3. A true pond with open water requires both a visible bounded shoreline or edge and an interior that visibly looks like open water.

4. A coherent basin shape alone is not enough.
5. If a feature appears mostly dry, muddy, vegetated, shallow, or only faintly damp, do not classify it as a pond with open water.
6. If a pond-shaped basin is visible but open water is not clearly visible, choose no for pond presence.
7. If a basin or stock-tank footprint is visible without clear open water, do not classify it as a pond.
8. If uncertain between water and shadow, vegetation, wet ground, mud, structure, or tank, choose no for pond presence.
9. Small dark rectangular features in developed or disturbed areas are often not ponds.
10. Tiny dark features may be houses, sheds, tanks, pads, containers, shadows, or other manmade features. Do not classify them as ponds unless open water is unmistakable.
11. For wildfire drafting, size matters. If a feature looks too small, too narrow, too shallow-looking, or operationally insignificant, do not mark it as draftable.
12. If a feature may contain water but appears too small to realistically support drafting operations, set likely_draftable to no or uncertain.
13. If uncertain, choose no for pond presence and explain why.
14. If a feature is uncertain, do not include a point.
15. If there are no visible ponds, return an empty list for pond_points.
16. In steep, rocky, alpine, or heavily textured terrain, dark enclosed features are often shadows, rock hollows, or terrain depressions rather than open water.
17. In rocky terrain, do not classify a feature as a pond unless the interior appears smooth and water-like, with a clearly bounded shoreline that is distinct from surrounding rock texture and shadow.
18. If a dark feature contains visible internal texture, irregular rock pattern, or tonal variation similar to surrounding rock or shadow, do not classify it as open water.
19. Small dark pockets embedded in broken rock or cliff-like terrain should usually be treated as shadow or rock unless open water is unmistakable.
20. If the feature could be explained by terrain shadow or rock geometry, choose no for pond presence.

Appendix B3. Required JSON response schema

Return JSON with exactly the following keys:

{
"is_there_surface_water": "<yes|no>",
"is_there_a_pond": "<yes|no>",
"road_or_access_nearby": "<yes|no|uncertain>",
"likely_draftable": "<yes|no|uncertain>",
"pond_count": ,
"evidence": "<90 words or fewer>",
"notes": "",
"pond_points": [
{
"label": "",
"confidence": <float from 0 to 1>,
"x": <float from 0 to 1>,
"y": <float from 0 to 1>
}

]
}

Rules for pond_points

- Coordinates must be normalized to the displayed crop:
    - x = column / image width
    - y = row / image height
- All coordinates must be between 0 and 1.
- Use one center point for each clearly visible pond or open-water feature.
- The point should be at the approximate center of the visible pond or open-water body.
- If a feature is uncertain, do not include a point.
- If there are no visible ponds, return an empty list.

Decision rules

- If there is no clearly visible open water, is_there_a_pond must be no.
- If a shoreline is not visible, is_there_a_pond should usually be no.
- If the interior does not look like open water, is_there_a_pond must be no.
- If the feature could reasonably be shadow, vegetation, mud, wet ground, structure, container, equipment, roof, or tank, is_there_a_pond must be no.
- If a pond-shaped basin or impoundment is visible but clear open water is not, is_there_a_pond must be no.
- If pond_count is 0, pond_points must be an empty list.
- If a pond appears too small or operationally insignificant for drafting, likely_draftable should be no or uncertain.
- Do not set likely_draftable to yes unless the feature appears plausibly large enough and operationally usable from the image.
- road_or_access_nearby refers only to what is visible in the image, such as a road, driveway, turnout, pull-off, or clear vehicle approach immediately adjacent to the pond.
- If a clearly visible road, driveway, turnout, or open vehicle-access area reaches or lies immediately adjacent to a clearly visible pond, set road_or_access_nearby to yes.
- If access is not clearly visible, set road_or_access_nearby to no or uncertain.
- A yes for road_or_access_nearby does not guarantee draftability by itself, but it is strong positive evidence.
- If there is a clearly visible pond with open water, road or access immediately adjacent to it, and the pond is not obviously tiny, prefer likely_draftable: yes.
- If there is a clearly visible pond with open water and road or access immediately adjacent, but the pond may be too small or operationally marginal, set likely_draftable to uncertain.
- If there is no visible access near the pond, likely_draftable should usually be no or uncertain.
- In rocky or mountainous terrain, darkness plus enclosure is not enough; require a smooth open-water appearance and a distinct shoreline.
- If a feature is small, dark, irregular, and embedded in rocky textured terrain, prefer is_there_a_pond: no unless open water is unmistakable.
- If confidence is not high that the feature is true open water, set pond_count to 0 and return no point.
- Be conservative, but do not ignore obvious direct road access when judging likely draftability.

The primary instruction was not to hallucinate features that were not visibly supported by the imagery.

Appendix B4. Model and request settings

The baseline workflow used OpenAI GPT-5.1, identified through the API alias gpt-5.1, with the temperature parameter set to 0. Each request contained one RGB NAIP image chip and required a structured JSON response. The baseline workflow used 500 m × 500 m image chips. The 1,000 m sampling-design sensitivity analysis used the same instructions and response schema, with the scene-width description changed to 1,000 m × 1,000 m.

If the initial model response could not be parsed as valid JSON, the image and original prompt were submitted a second time with the following additional instruction:

"Your previous response was not valid JSON. Return only valid JSON matching the exact schema requested. Do not include markdown, prose, or comments."

Claude Sonnet 4.6 received the same baseline 500 m image chips, classification definitions, interpretation criteria, and structured output requirements during the model-sensitivity analysis.